Constraint-driven generation and forming-program derivation for hairpin end windings

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Abstract. The end-winding shape of a hairpin conductor determines copper mass, resistance and the clearance between neighbouring pins, yet it is usually fixed by hand and checked afterwards. We describe a workflow in which every candidate shape is a complete CAD model, all geometric requirements are evaluated as signed constraints, and a surrogate-model-based search proposes the next shapes to build. For a 36-slot, six-layer stator with three insertion baskets, about 96 shape evaluations per basket yield a trade-off between conductor length and clearance. The selected pins have a common end-winding height within 0.5 mm. A second step fits a forming program of feed, twist, head rotation and bend to the optimized three-dimensional centreline. Including an elastic–plastic springback model, the programs of 20 to 21 bends reproduce the target curves within 0.9 to 1.2 mm, compared with about 9 mm if springback is ignored. Material and tool values are assumptions; no experimental validation has been carried out.

1 Introduction

Hairpin windings use pre-formed rectangular conductors and reach slot fill factors of 80–85 % [1]. Their production is commonly divided into shaping, assembly, twisting and contacting, and the shaping of the conductor is regarded as a critical step [1]. The conductor ends outside the stator stack, the end winding, carry no torque but add resistive loss, mass and installation space; reducing the end-winding length is one of the levers for a lower winding resistance listed in [1]. Their shape is limited by competing requirements: a minimum clearance to the neighbouring conductors, a minimum bend radius of the rectangular wire, the available installation space, and a feasible insertion sequence in which groups of pins (baskets) are joined one after the other.

In practice the shape is drawn, checked and corrected iteratively. Two difficulties make this slow. First, the requirements are coupled: a tighter bend shortens the conductor but reduces the clearance to the neighbour, while a larger bend radius raises the height. Second, a formally correct shape can be hard to form, because a continuous curve of varying curvature has to be approximated by a finite sequence of machine motions, and the wire springs back after each bend. We therefore treat shape generation and manufacturing as two consecutive modules operating on the same pin (Fig. 1) and ask how short a pin can be made under all requirements, and how closely a bend program can reproduce it.

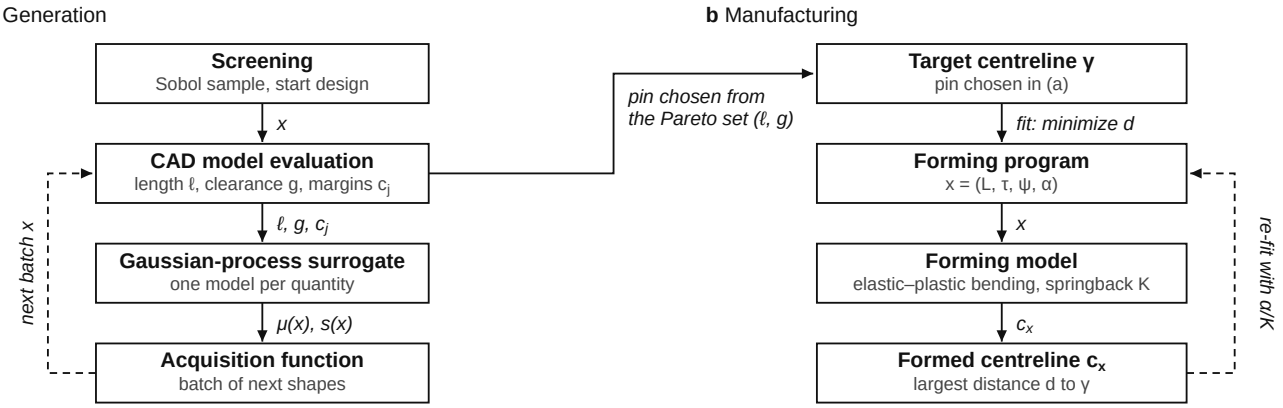


Figure 1. Workflow of the two modules. (a) Generation: shapes are evaluated on CAD models; a Gaussian-process surrogate and an acquisition function propose the next batch (dashed: feedback). (b) Manufacturing: a forming program is fitted to the chosen centreline; a forming model including springback predicts the formed pin, and the program is re-fitted until the largest distance *d* is minimal.

2 Methods

Problem definition. The test case is a stator with 36 slots, six conductor layers and a rectangular conductor of 4.5 mm × 3.4 mm (Table 1). The winding is divided into three baskets of 36 pins each. The end-winding shape of one pin is described by a small set of shape parameters that define the three-dimensional centreline of the conductor, from which the conductor body is built.

Table 1. Test case and requirements (all treated as constraints).

Item	Value	Item	Value
Stator bore / outer diameter	136 / 219 mm	Slots / layers / baskets	36 / 6 / 3
Stack length	156.1 mm	Conductor (width × thickness)	4.5 × 3.4 mm
Installation space (ring)	Ø132–219 mm, height ≤ 70 mm	Minimum bend radius	related to the cross-section
Minimum clearance (CAD)	0.1 mm	Insertion	basket by basket

2.1 Evaluation and problem statement

Every candidate x in the scaled parameter cube $[0,1]^d$ is constructed as a parametric CAD model and measured, so that no requirement

rests on an analytical estimate of the final geometry. The measured quantities are the conductor length $\ell(x)$, the smallest distance to the neighbouring conductors $g(x)$, an angular margin, the bend radius relative to the cross-section, and the margin to the installation space. Each requirement is expressed as a signed margin $c_j(x)$ with $c_j(x) \geq 0$ for a satisfied requirement, so that a violating candidate still carries information about how far it is off. Parameter sets that cannot describe a geometrically admissible shape are identified beforehand, so that no evaluation is spent on them. The problem is

$$x^{*} = \arg\min \ell(x) \quad \text{subject to} \quad c_j(x) \geq 0,\ j = 1 \ldots m,$$

and, for the trade-off, the pair $(\ell, -g)$ is minimized in the Pareto sense: a pin x dominates x' if it is no worse in both quantities and better in one.

2.2 Screening and sensitivity analysis

The search starts from an initial design and a space-filling sample of admissible shapes (scrambled Sobol sequence). A Gaussian-process regression with a Matérn 5/2 kernel and automatic relevance determination is fitted to each measured quantity,

$$k(x,x') = \sigma^2 (1 + \sqrt{5}r + 5r^2/3) \exp(-\sqrt{5}r), \quad r^2 = \sum_i (x_i - x'_i)^2 / \lambda_i^2,$$

with one length scale λ_i per parameter. The hyperparameters follow from maximizing the marginal likelihood, and the model quality is checked by cross-validation. On the model mean, first-order and total Sobol indices, $S_i = \mathrm{Var}(\mathrm{E}[Y|x_i])/\mathrm{Var}(Y)$ and $S_{Ti} = \mathrm{E}[\mathrm{Var}(Y|x_{-i})]/\mathrm{Var}(Y)$, quantify the influence of each parameter on each quantity. Parameters without effect on any target are fixed, which reduces the dimension of the subsequent search. The indices are a screening and not a proof, because the surrogate extrapolates in regions of the parameter space where no admissible shape exists.

2.3 Bayesian optimization

The reduced problem is solved by Bayesian optimization in the sense of [2]: a probabilistic surrogate model together with an acquisition function that uses the model uncertainty, a setting suited to costly evaluations without derivatives [2]. One Gaussian process, with the kernel above, is fitted per objective and per constraint. Given the data, each process yields a posterior mean $\mu(x)$ and variance $s^2(x)$. For a single objective the expected improvement over the best feasible value f^{*} is

$$\mathrm{EI}(x) = \mathrm{E}[\ \max(f^{*} - f(x), 0)\ ],$$

which is large where either the predicted value is low or the uncertainty is high. Because several candidates are proposed at once and the observations are modelled with a noise term, a batch form of the expected improvement is used. Constraints that are only known from evaluation enter through their own surrogates and weight the improvement by feasibility, in the spirit of the probability-weighted expected improvement described in [2]. The candidates of a round are evaluated in parallel and the surrogates are refitted.

For the trade-off between length and clearance the objectives are normalized and combined by an augmented Chebyshev scalarization with a weight vector w drawn at random from the simplex for every proposal,

$$s_w(y) = \max_i w_i\, y_i + \rho \sum_i w_i\, y_i,$$

as in ParEGO, which randomly scalarizes the objectives and applies expected improvement [3]. Different weights steer successive proposals to different parts of the front, which is read from all evaluated shapes. Hypervolume-based acquisition, which maximizes the expected gain of the dominated area and is available for parallel and constrained problems, is the alternative treated in [3]. The result is not one pin but the set of non-dominated pins; the final choice is an engineering decision. The three baskets are optimized in sequence, and the selected pins are chosen so that their end-winding heights agree.

2.4 Forming program

The selected centreline $\gamma(s)$ is approximated by a sequence of motions of a rotary-draw forming process. A local frame (t, u, b) travels along the wire; step i consists of a feed L_i along t, a twist τ_i about t, a rotation ψ_i of the bending plane about t, and a bend of angle α_i on a circular arc of the tool radius R. The centreline c_x is the chain of these rigid motions for the parameter vector $x = (L_i, \tau_i, \psi_i, \alpha_i)$. The parameters are fitted to minimize the largest two-sided distance to the target,

$$\min_x \max_s \mathrm{dist}(c_x(s), \gamma) \quad \text{subject to} \quad 0 \leq \alpha_i \leq \alpha_{\max}, \quad |\tau_i| \leq \tau_{\max},$$

for a fixed tool radius R that is checked against a minimum radius. The maximum norm is non-smooth. It is approached through smooth p-norm approximations and solved with a bounded nonlinear least-squares method from several starting programs, with the number of bends varied.

Springback is described with a power-law hardening material, $\sigma = \sigma_y (\varepsilon/\varepsilon_y)^n$ beyond the yield strain ε_y. For a rectangular cross-section bent to curvature κ the strain varies linearly over the thickness, $\varepsilon = \kappa z$, and the bending moment is $M(\kappa) = \int \sigma(\kappa z)\, z\, \mathrm{d}A$. Unloading is elastic, so the curvature recovers by $\Delta\kappa = M/(EI)$ and the bent angle is reduced by a factor $K = 1 - \Delta\kappa/\kappa$ (close to 0.98 for the assumed copper data). The program therefore commands α_i/K_i and is re-fitted as a whole, because a correction of one bend shifts every subsequent one.

3 Results

Generation. For basket 1, 96 shapes were built and measured: the start design, 15 screening shapes and 80 shapes proposed by the learning search. 42 of them satisfy all requirements (Fig. 2a). The feasible shapes span end-winding lengths from 174 to 201 mm and clearances from 0.2 to 2.4 mm; the non-dominated pins form a staircase in which about 24 mm of additional length buys roughly 1.6 mm of clearance (Fig. 2b). None of the 15 screening shapes was feasible, whereas 41 of the 80 shapes proposed by the learning search were. Selecting one pin per basket under a common end-winding height gave heights of 66.3, 66.1 and 66.6 mm (Table 2). The

complete stator assembled from the three pins is shown in Fig. 4 (Section 4).

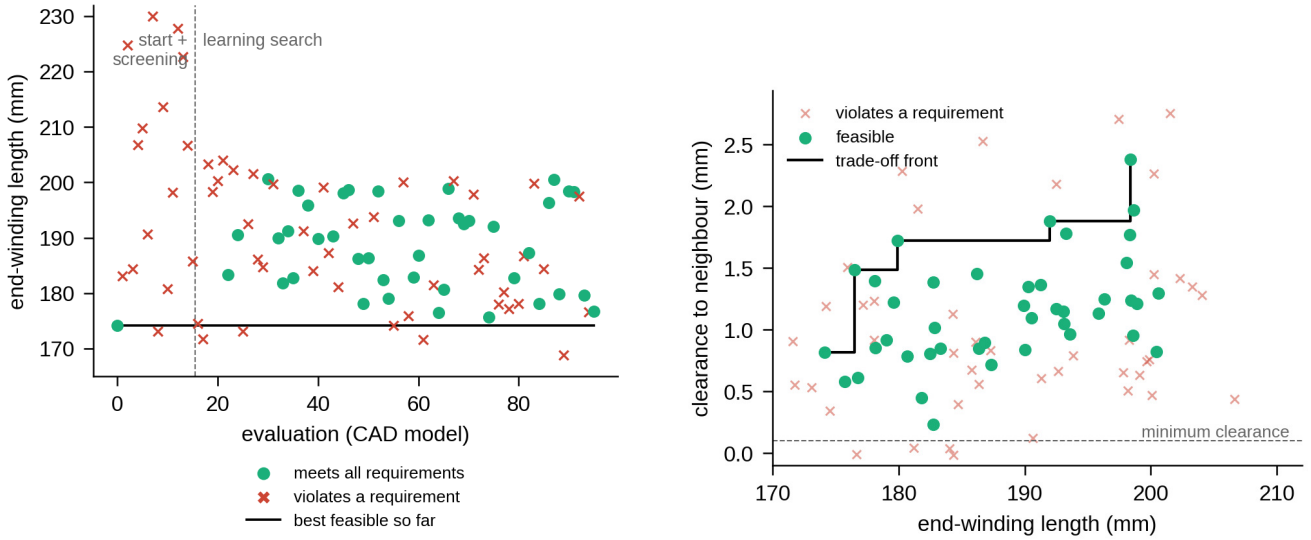


Figure 2. Generation run for basket 1. (a) The 96 evaluations in the order they were built; the line is the shortest feasible length so far. (b) End-winding length versus clearance; the black line is the non-dominated front.

Manufacturing. For the three selected pins, forming programs of 20, 21 and 20 bends were obtained with bend angles up to 32° and twist of at most 20° per straight piece (Table 2). For the fitted programs the largest distance between the formed and the target centreline is 0.89, 1.16 and 1.04 mm (without springback). If a wire with the assumed material springs back and the programs are run uncompensated, the distance accumulates over the bends to 8.7–9.1 mm; with over-bending and re-fitting the model gives 0.9–1.1 mm (Table 2). The commanded angles differ from the design angles by a few per cent, with a larger relative change for small bends. Fig. 3 shows the program of the pin of basket 1; the other two pins have the same structure. The first 160 mm of the wire are fed straight, the bending is concentrated between 160 and 295 mm, and several bends around 230 mm follow each other with practically no feed. The head rotation covers the range from −180° to +91°, and the twist stays within the limit of 20° per piece. The distance at the end of the pin stays below 0.2 mm in all three cases.

Table 2. Results per basket: end-winding length and common height, number of bends, largest bend angle and twist, and the largest distance to the target centreline for the fitted program (Fit), with uncompensated springback (Uncomp.) and with compensation (Comp.). Manufacturing values are model results for assumed material and tool data.

Basket	Length (mm)	Height (mm)	Bends	Max. bend (°)	Max. twist (°)	Fit (mm)	Uncomp. (mm)	Comp. (mm)
1	196.3	66.3	20	29.7	18.5	0.89	9.1	0.90
2	191.1	66.1	21	22.9	20.0	1.16	8.8	1.14
3	181.8	66.6	20	31.6	19.9	1.04	8.7	1.07

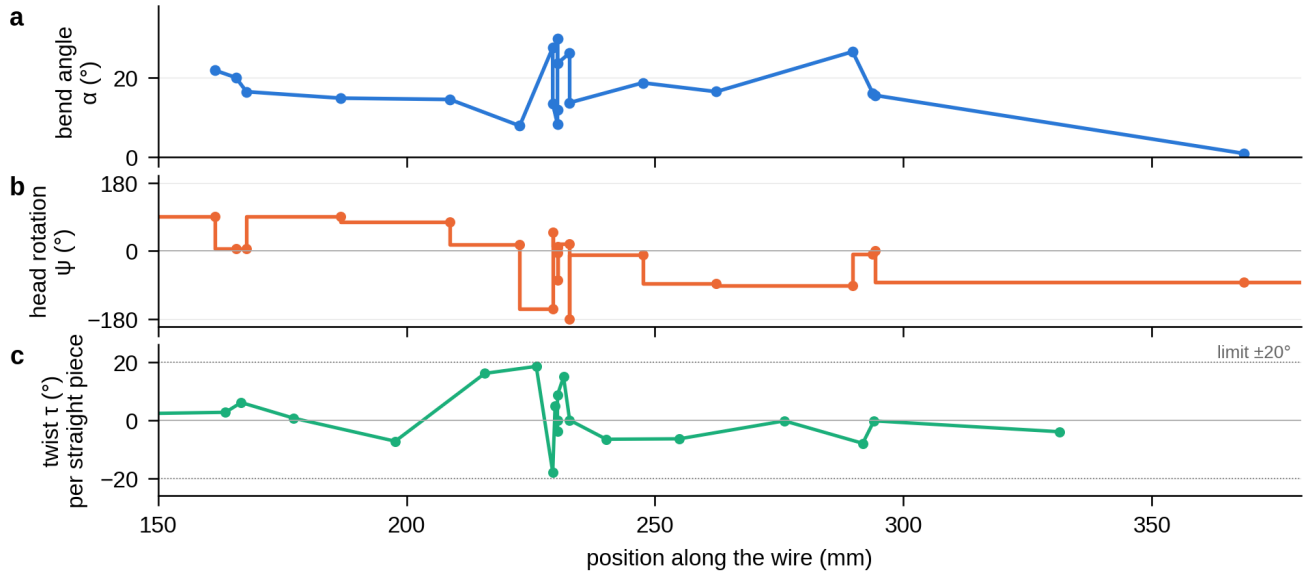


Figure 3. Forming program of the pin of basket 1 along the wire, from the first feed (position 0) to the end of the pin. (a) Designed bend angle of each bend; the angles commanded after springback compensation differ from it by less than 1°. (b) Rotation of the bending head. (c) Twist per straight piece with its limit.

4 Discussion

Two points deserve comment. The first concerns the coupling of the requirements. The front in Fig. 2b is not smooth: a large part of the shape space is infeasible (54 of the 96 evaluations), and feasibility is not a simple function of any single parameter. This is the reason for evaluating the requirements on the actual CAD geometry rather than on a simplified length model; the crossings of neighbouring pins in the end winding, where the clearance requirements act, are visible in the sections of Fig. 4. The second concerns the cost of springback. The deviation of about 9 mm without compensation is not a property of one bend but of the accumulation over 20 bends, and a correction that is applied per bend does not remove it; the whole program has to be re-fitted.

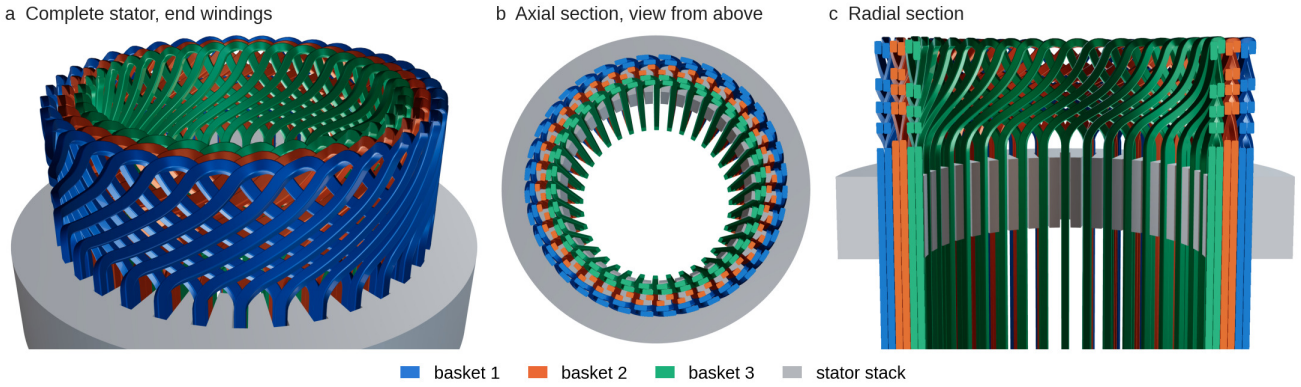


Figure 4. Stator assembled from the three selected pins (36 slots, six layers, 108 hairpins). (a) End windings of all baskets. (b) Axial section, viewed from above: the baskets occupy separate radial bands. (c) Radial section: the pins of adjacent slots cross in the end winding. Rendering of the CAD-derived geometry.

The choice of a surrogate-based search follows from the cost of one evaluation. Evolutionary algorithms such as NSGA-II are robust multi-objective optimizers but need many function evaluations, whereas Bayesian optimization is the more sample-efficient alternative when each evaluation is expensive [3]. Its known restriction to problems of moderate dimension [2] is the reason for the screening step, which reduces the number of free parameters before the search. The Gaussian-process length scales and the Sobol indices both rest on a smooth surrogate of a response that is not smooth near the boundary of the admissible region; the sensitivities are therefore indicative.

The limitations are substantial. In the basket-1 run the shortest feasible shape coincides with the start design, so the search contributed mainly the characterization of the trade-off and the pins for the common height rather than a shorter conductor at minimum clearance. The clearance limit of 0.1 mm is a geometric limit of the model and not an insulation specification. The manufacturing results are based on assumed material and tool data, are not checked for collisions of tool and neighbouring pins, and carry no experimental validation; the stated distances are therefore model values, not measured accuracies. The fit is a local optimum of a non-convex problem, and repeated fits with different bend counts scatter by several millimetres, so only the best result over several bend counts should be read as an achievable value.

5 Conclusions

Treating each candidate as a verified CAD model and each requirement as a signed constraint allows a surrogate-based search to explore the end-winding shape with fewer than 100 evaluations per basket. The same centreline can be turned into a forming program whose largest deviation is about 1 mm in the model once springback is included in the fit, whereas ignoring it causes about 9 mm.

6 Outlook

The next steps are a collision check of the forming program against tools and neighbouring pins, calibration of the material model with tensile and bending tests, and a measurement of formed pins against their target curves. On the winding side the approach is to be extended to the twist side of the stator, to busbars and to special pins. Measurement data and manufacturing experience from other groups would be valuable for this; contact is possible under the address given above.

Availability. Screenshots of the two applications and further information on the project are available at https://fronteerlab.com; access to the applications is available on request. All values in this report are model results for the assumptions stated in the text.

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